System Design Specifications – Team 2

Introduction/Management Summary

Purpose of the Document:

The purpose of the Systems Design Specification (SDS) is to describe how the new Performance Management and Salary System will be designed, built, and operated. It provides guidance to programmers, infrastructure teams, security analysts, and IT management so they can develop, deploy, and maintain the system effectively.

Brief Overview of the project:

Federated Insurance is replacing its outdated performance and salary system with a modern, automated solution. The new system will improve efficiency, reduce user error, integrate with existing HR systems, and support future growth. It is expected to save time and reduce operational costs across the company.

High-Level scope:

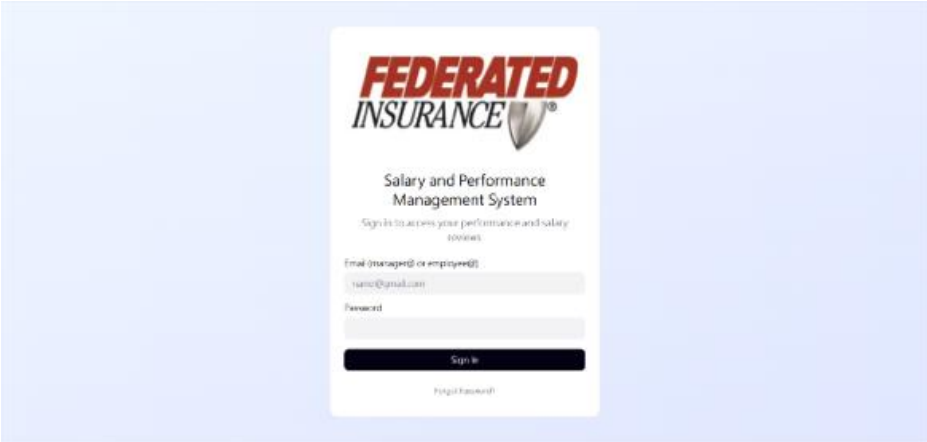
- **In Scope:**
  - Performance management functionality
  - Salary reporting and projections
  - Employee access to view their actual salary and performance data
  - Integration with existing HR and payroll systems
  - Automated notifications of salary changes, reviews, and IAPs
  - Employee performance reviews (annual, quarterly, on-demand)
  - MERIT-based salary adjustments
  - Role-based access and security controls
- **Out of Scope:**
  - Bonuses
  - Compa-Ratio calculations
  - Legacy “A, B, C” rating system will not be supported
  - Doesn’t replace HR modules (benefits, payroll, etc.)

Assumptions the System Should be Based on:

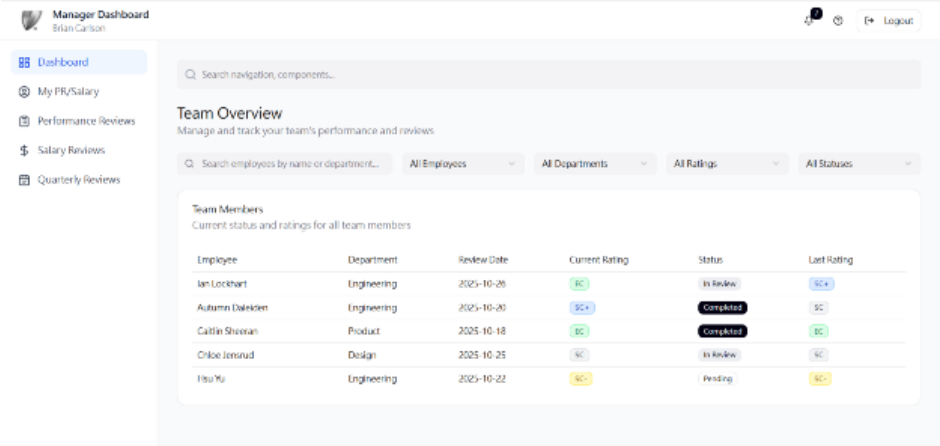
The system should be designed based on several key assumptions. The company is expected to grow by 5% in employee size each year, and the system will be used heavily and continuously after deployment. While the schedule is tight and may create internal conflicts, management fully supports the project. Training will be required for all employees, which is approximately 3,000 individuals. A prototype is not necessary due to the use of Agile methodology. The system must notify employees of salary changes at least three days before they take effect, and integration

with the existing HR system is required. Finally, the necessary hardware, software, and network resources are already in place to support development.

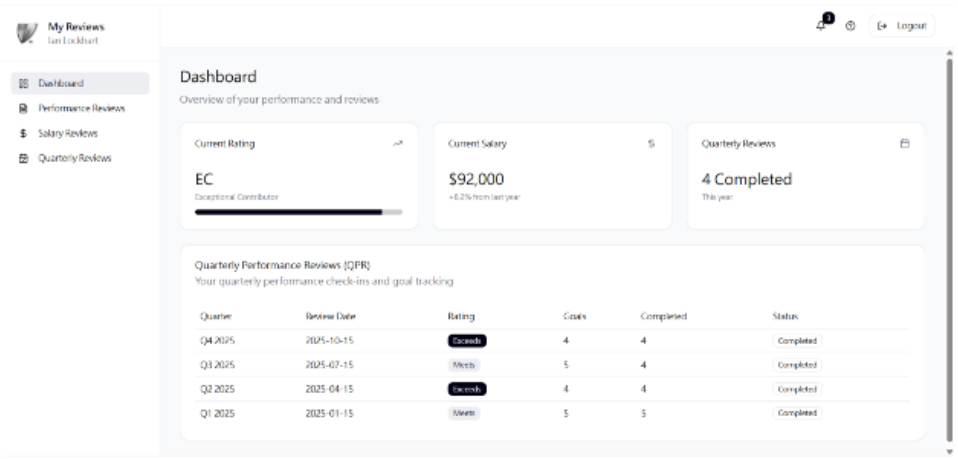
System Components:



- This is the log-in screen. Users will enter in their email and then a designated password to be logged into their account.



- This is the Manager landing page. You will see the first page after logging in as a manager. This screen includes as overview of all team members, the department they are in, the review date, the employees' current rating, status of their current review, and their last rating.



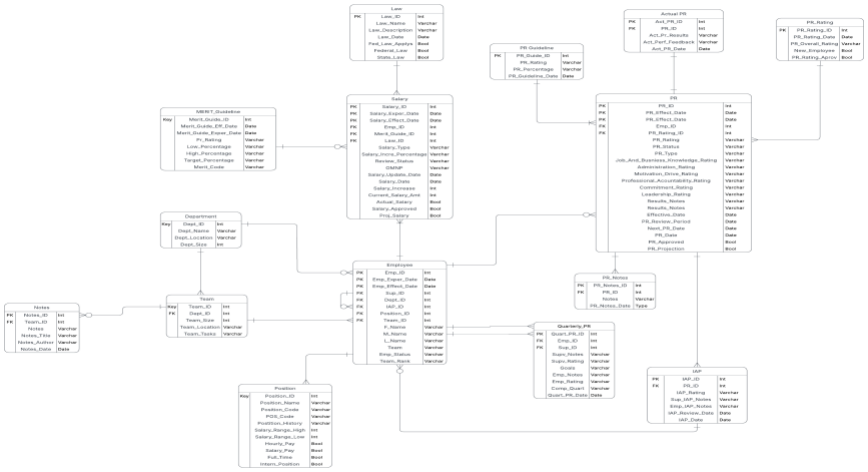
- This is the staff landing page. You will see the first page after logging in as an employee. This screen includes the staff's current rating, current salary and how many Quarterly reviews have been completed. Below it also includes which quarter and year the review was in, the date of the review, the rating that was received from the review, how many goals they had along with the amount they completed, and the status of the review.

System Inputs:

- Performance evaluations
- Salary projections and actuals
- Employee data (imported or received from HR system)
- Manager updates
- System-generated notifications

System Outputs

- Reports on actual and projected salaries
- Reports on actual and projected performance reviews
- Performance summaries
- Notifications of salary changes (required)
- Internal dashboards
- Data provided to payroll/HR systems
- Current salary with percentage increase



- Above is the entity relationship diagram of the system. It describes the major data entities and their relationship.

Integration and Network Specifications:

The system shall integrate with payroll and existing employee management systems to ensure seamless data exchange and maintain consistent employee information across platforms. It shall validate all data traffic, and in the event of communication failures, display an error message and log the incident. The system must support a minimum of 3,500 current employees and be scalable to accommodate an expected 5% annual growth without the loss of quality or the need for major changes. The system shall retain employee performance, salary, and workflow data for a total of five years, including the current active year plus the preceding four years. Data older than five years shall be archived or securely deleted according to organizational policy. Additionally, the system shall be accessible across multiple platforms, offering a user-friendly, responsive interface with full functionality for all authorized users.

Application detailed design:

Performance Review Module:

The system enables authorized users to create, configure, and complete annual, quarterly, and on-demand performance reviews. It allows employees with proper authorization to access quarterly reviews, leave comments, and provide signatures. The system supports quarterly performance reviews in which employees record accomplishments, set goals, and provide self-assessments, all of which roll up into the annual reviews. Additionally, it tracks scheduled, completed, and missing reviews, sending notifications for follow-up when data is incomplete.

Merit and Salary Module:

The system enables HR staff to maintain and update salary ranges, performance review guidelines, and merit charts on an annual basis. HR merit increase structures and automatically calculates salary adjustments based on MERIT ratings and the predefined percentages in the MERIT chart. The system ensures that managers and supervisors remain within budget when adjusting salaries outside MERIT chart targets by displaying errors or requesting approval if limits are exceeded. Additionally, it notifies employees of salary changes three days in advance and applies these changes on a Sunday.

Workflow and Approval Module:

The system provides workflow-based approval routing, moving requests from employees to manager, supervisor, director, executive director, and finally the CIO. It notifies employees of required approvals, logs all decisions, and informs the requestor of whether their request was approved or denied. If a request is denied, the system terminates the workflow.

Security and Access Module:

The system enforces role-based access control, ensuring that data is accessible only to authorized users. It prevents users from viewing information outside their reporting hierarchy, displaying and logging errors if such attempts occur. Access is permitted through authorized company devices and verified personal devices using two-factor authentication. All salary and performance data are encrypted both in transit and at rest, with failed encryption attempts logged and access denied.

Interface with Purchased Software Packages:

The system shall integrate with payroll and employee management systems for data exchange, using secure validation protocols.

System Environment

Constraints and Conditions:

The system must support at least 3,500 employees with 5% annual growth and handle heavy activity during the November–January projection and budgeting period. It must meet strict timelines, including notifying employees of salary changes three days before they take effect and applying all salary updates on Sundays. The system must generate electronic performance and salary reports, store all PR and salary histories, and maintain fast, reliable response times. It must also support continuous updates to performance reviews, salary projections, merit guidelines, and HR-related data.

The system must run on secure company devices and authorized personal devices, requiring reliable processing power, adequate storage, and support for multi-platform access. It must integrate with existing HR, payroll, and employee management systems, validating data

exchanges and logging any failures. Database updates must occur regularly for performance reviews, salary changes, and merit chart adjustments.

Strict security requirements must be enforced through role-based access control, ensuring users can only view information within their reporting structure. Staff may view only their own actual data, and projections must remain restricted. All sensitive data must be encrypted in transit and at rest. The system must use two-factor authentication and log all access, changes, approvals, and exceptions.

The system requires standard 120V AC power input for all workstation devices and server equipment, with compatible grounded connectors to ensure stable operation. Hardware components should use Cat6 or higher Ethernet cabling, with typical cable lengths ranging from 3 to 25 feet depending on workstation and network rack placement. User interfaces will operate through standard computer workstations, supporting a minimum monitor resolution of 1920×1080 to properly display dashboards, forms, and reports. Devices must include sufficient local storage for temporary caching and processing, while servers must include scalable storage to house performance data, salary records, logs, and HR integration files. Processor requirements include modern multi-core CPUs capable of supporting continuous multi-user access and real-time calculations.

Communication Detailed Design

Data Exchange Requirements:

The communication layer must support the transfer of performance data, salary information, workflow approvals, and system notifications. All exchanged information must remain encrypted in transit, with structured formats ensuring compatibility across modules and integrated systems.

Error Handling, Backup, and Recovery:

All errors will be logged automatically, with notifications retried when transmission failures occur. Backup and recovery processes include daily incremental backups, weekly full backups, and long-term file retention for compliance. These processes ensure the system can restore data quickly and maintain continuity.

Database Update Requirements:

Database updates occur whenever performance reviews, salary projections, merit charts, HR guidelines, or employee information changes. Missing or incomplete data triggers automated alerts. Tables, views, and records update continuously throughout the workday as users input reviews, projections, and approvals.

Support Processing:

Support processing includes system startup sequencing, cleanup of outdated log files, and secure long-term retention of critical records. Backup services protect data integrity, and recovery procedures allow the system to restore operations in the event of a failure.

System Update Frequency:

The system receives new data daily from HR imports, performance review activity, workflow decisions, and salary calculations. These frequent updates generate steady transaction volume that the communication layer must handle efficiently to maintain system responsiveness.